Digit-Sum Reachability on the Lattice: An Experimental Expedition from Brute Force to a Two-Line Criterion

Adam Benzan*

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Abstract

Starting from $(0, 0)$, a walker moves on $\mathbb{Z}^2$ through cells satisfying $\mathsf{s}(|x|)+\mathsf{s}(|y|) \le N$, where s is the digit sum (base 10 unless noted). We chronicle an experimental program on the reachable set: exact counts and the fractal geometry of the region; a Cantor-like staircase in N with cliffs at multiples of 9; an exact closed form in base 2, $Q(N) = \binom{2N+3}{N}$; a carry-free self-similar ("gastank") decomposition giving a polynomial-time exact solver; the asymptotic law reachable/allowed in box $\sim c/\sqrt{N}$ with $c = 3.84717(5)$ (refuting a tempting $\sqrt{15}$); and finally an exact two-line reachability criterion, proved in a companion paper [1], which subsumes every structure met along the way and reduces all counting to a one-dimensional walk recursion. We keep the narrative in expedition order, including two instructive false summits.

1 The problem

A cell $(x, y) \in \mathbb{Z}^2$ is *allowed* iff $\mathsf{s}(|x|) + \mathsf{s}(|y|) \le N$; the *reachable set* is the 4-connected component of the origin within the allowed set. Write $f(N)$ for its size and $Q(N)$ for the size of its closed-quadrant part. The canonical instance is $N = 19$:

$$f(19) = 102{,}485, \qquad \text{bounding box } [-298, 298]^2,$$

the box ending exactly at the first *axis cliff*: $\mathsf{s}(299) = 20$ makes the entire column $|x| = 299$ forbidden. The region is 4-fold symmetric, diamond-like at large scale, pierced by rectangular "death zones" and nested self-similar triangular voids (Figure 1).

Allowed versus reachable

Each row y of the allowed set is the one-dimensional set $\{x : \mathsf{s}(|x|) \le N - \mathsf{s}(|y|)\}$, so the allowed set factors through a family of 1-D threshold sets. Counting allowed cells in the bounding box at $N = 19$ gives 166,973, versus 102,485 reachable: a gap of 64,488 cells (38.6%) that satisfy the constraint but are cut off behind carry cliffs — *islands*. The canonical island: $(300, 0)$ is allowed ($\mathsf{s} = 3$) but hidden behind the forbidden column $x = 299$. Figure 2 shows the 1-D digit-sum signal whose threshold crossings generate the cliffs.

*Joint exploration with Claude (Anthropic). Scripts, exact datasets, and the companion proof paper live in the project repository.

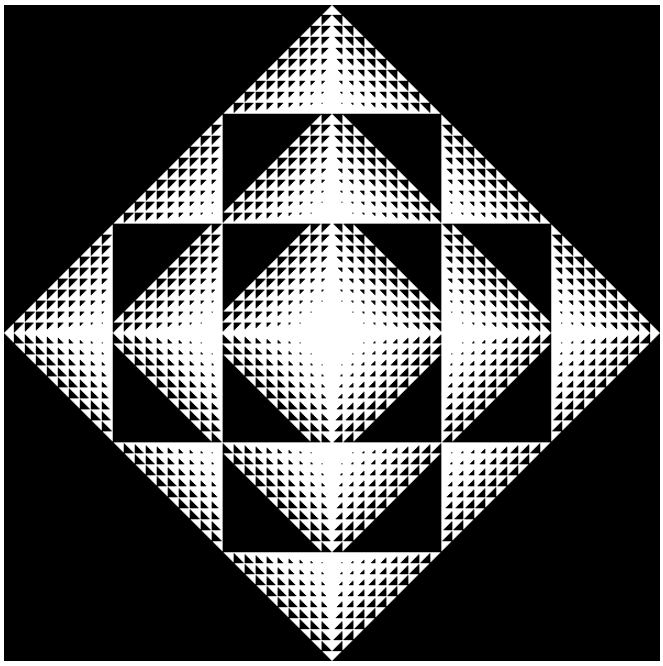


Figure 1: The reachable set at $N = 19$ (102,485 cells, box $[-298, 298]^2$): diamond envelope, rectangular death zones, and the nested "tri-force" triangular voids from the self-similarity of digit sums at powers of ten.

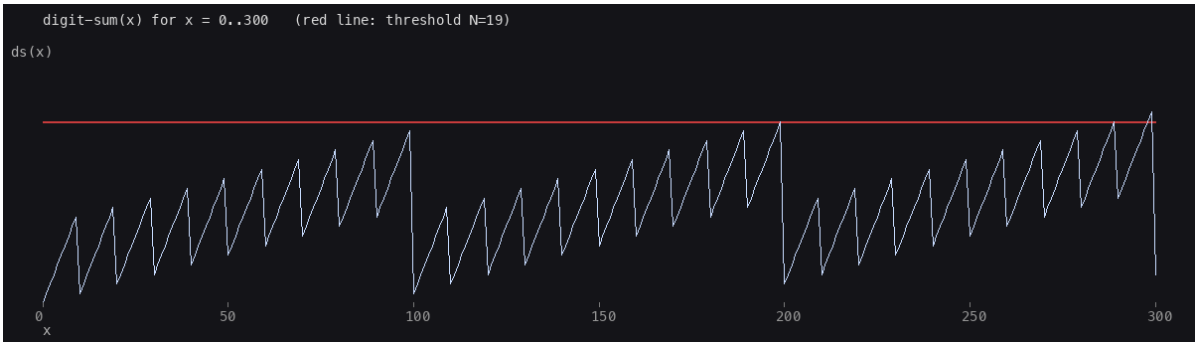


Figure 2: The digit-sum signal $\mathsf{s}(x)$ for $x = 0..300$ against the threshold $N = 19$ (red). Reachability along the axis ends at the first crossing, $\mathsf{s}(299) = 20$; each row of the 2-D problem sees this same signal at a reduced threshold.

2 The staircase in N

Exact counts as N grows show a descending, Cantor-like pattern in the ratio $f(N)/\text{allowed-in-box}$, with cliffs exactly at $N \in \{9, 18, 27, 36\}$ — the thresholds at which the box first jumps a decade.

N	$f(N)$	allowed (box)	ratio	box
8	145	145	1.0000	±8
9	505	649	0.7781	±18
17	14,785	18,265	0.8095	±98
18	47,905	73,885	0.6484	±198
19	102,485	166,973	0.6138	±298
26	1,309,537	1,885,501	0.6945	±998
27	4,216,357	7,552,009	0.5583	±1998
35	117,109,345	190,327,945	0.6153	±9998
36	377,329,873	761,526,445	0.4955	±19998

Within each decade the topology is fixed and holes fill in monotonically; each new decade adds an outer, rescaled copy of the previous decade's pattern (Figures 3 and 4). The ratio minimum falls two steps *after* each cliff ($N = 11, 20, 29$), not at it.

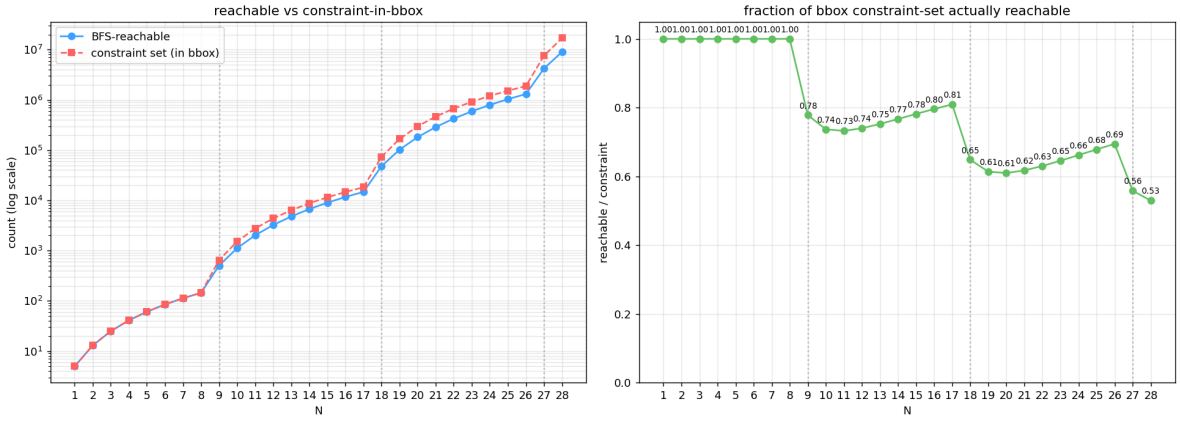


Figure 3: The Cantor-like staircase: reachable count, allowed-in-box count, and their ratio for $N \leq 28$, with cliffs at $N = 9, 18, 27$ where the bounding box jumps a decade.

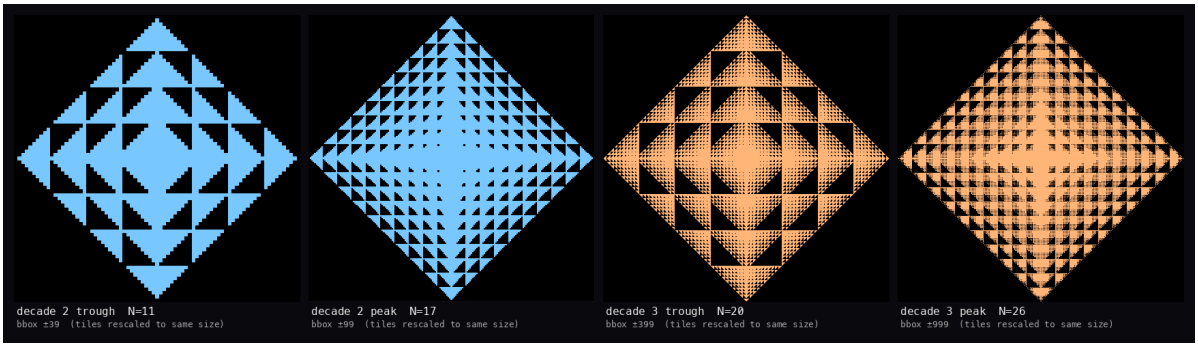


Figure 4: First-of-decade (trough) versus last-of-decade (peak) shapes for decades 2 and 3. The $N = 11$ trough is the $N = 20$ trough rescaled; the recursion made visible.

3 The monotone lemma

A BFS-depth experiment found *detour* := geodesic − (|x| + |y|) to vanish on every reachable cell: whatever is reachable at all is reachable by a monotone staircase. Equivalently, in the quadrant,

$$R(x, y) = \text{allowed}(x, y) \wedge \big(R(x{-}1, y) \vee R(x, y{-}1)\big)$$

reproduces full undirected reachability. Verified for $N \le 22$ (base 10) and for bases 2–5; later *proved* as a corollary of the criterion of §8. Connectivity failures are always total (islands), never partial (no cell requires a detour); see Figure 5. Practically, the lemma converts BFS into a vectorized row sweep, ∼100× faster — the workhorse for everything below.

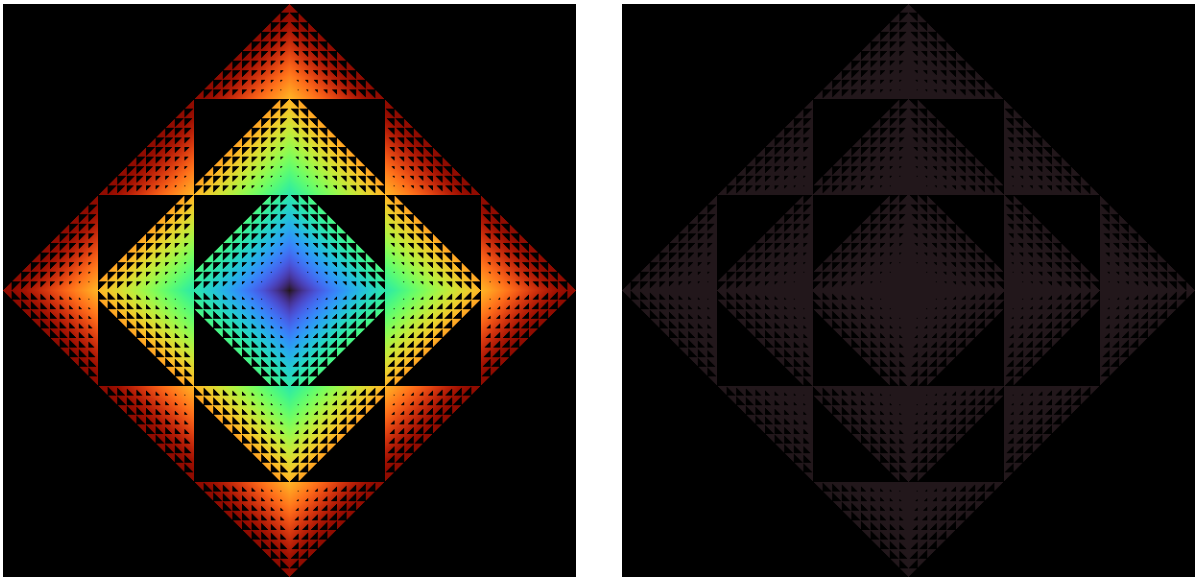


Figure 5: Left: cells colored by geodesic (BFS) distance from the origin at $N = 19$. Right: detour = geodesic − (|x| + |y|), identically zero — the observation that became the monotone lemma and, eventually, the criterion.

4 Base 2: an exact closed form

In base 2 the quadrant count is a single binomial coefficient:

$$Q_2(N) = \binom{2N+3}{N}, \qquad f_2(N) = 4\binom{2N+3}{N} - 2^{N+3} + 5,$$

first verified for $N \le 15$ (the $N = 15$ value *predicted before* a 4.3×10^9-cell sweep confirmed it), later *proved* from the criterion: interleaving the bits makes reachability a ±1-path condition whose even-checkpoint constraint collapses, by parity, to the uniform barrier "never reach +2", and reflection at level 2 telescopes the count to $\binom{2k}{k-1} + \binom{2k}{k-2}$ — the allowed cells of the top two weight layers, with an explicit *surfacing map* bijection (complement prefixes up to first hits of level −2). The allowed-set side is classical, $\frac{1}{2}\big(4^{N+1} - \binom{2N+2}{N+1}\big)$, so the island count is closed-form by subtraction, and

$$\frac{Q_2(N)}{\text{allowed}} \longrightarrow \frac{4}{\sqrt{\pi N}}.$$

The ratio tends to *zero*: there is no finite asymptotic density. Q_2 is OEIS A002054 (ballot numbers), pointing to a lattice-path bijection; the base-10 sequence $f(N)$ is not in OEIS. Figure 6 compares the region across bases: base 2 is densest and nearest a solid diamond (carries at every step weave fine cliffs), while larger bases have rarer but more violent carries, punching larger holes.

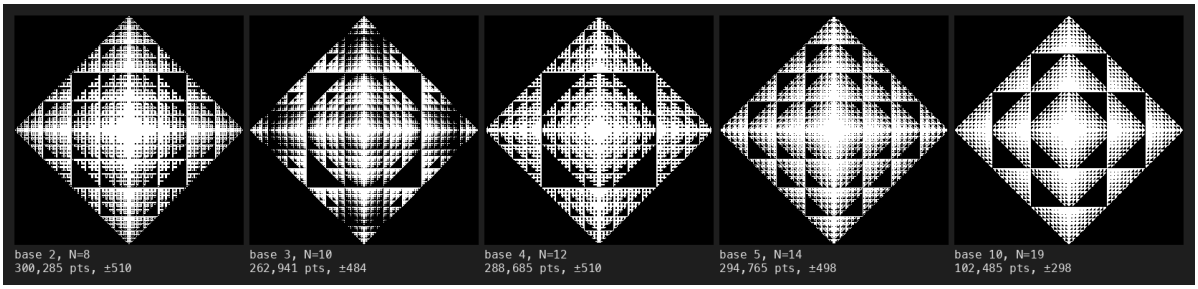


Figure 6: Reachable regions in bases $2, 3, 4, 5, 10$, with N chosen so each has axis reach $\sim \pm$ 250–500. Base 10 is the noisiest member of the family.

5 Gastanks: the carry-free decomposition

Definition 1 (Tank, wall, weir, wetting front)**.** A *tank* $T_k(K)$ is a block $[0, 10^k)^2$ in local coordinates with budget K: local cell (u, v) allowed iff $\mathsf{s}(u) + \mathsf{s}(v) \le K$. Since digit sums are carry-free inside a block, the child at digit position (i, j) is exactly $T_{k-1}(K - i - j)$: budgets step down by one per digit, with no corrections, ever. A *wall* is the trailing-nines seam between sibling tanks (digit-sum cost $9s$ at sibling scale s); carries *are* walls. A *weir* is the passable part of a wall, $\{v : \mathsf{s}(v) \le K_{\text{left}} - 9s\}$ — itself a 1-D threshold set one scale down. A *wetting front* is the set of edge cells through which flow passes; fronts are Cantor-dust-like with up to $\sim 9^{k-1}$ intervals.

Equivalently, in *terrain* coordinates (elevation = $\mathsf{s}(x) + \mathsf{s}(y)$, waterline = N): walls are ridges, weirs are notches dipping below the waterline, islands are submerged basins sealed off by ridges that never dip. All four terms appear together in a single real strip of the N=19 plane in Figure 7.

Flooding tanks recursively — passing fronts over weirs, with fronts hash-consed as interned 10-tuples of child fronts (a DAG; never expanded) — gives an exact solver whose distinct-state count grows only polynomially ($\sim N^{3.3}$). Benchmarks (cold start, cross-checked outputs):

N	BFS (set)	BFS (grid)	monotone DP	tanks	$f(N)$
26	1.31 s	1.19 s	0.009 s	0.002 s	1.3×10^6
28	—	8.55 s	0.053 s	0.003 s	9.0×10^6
36	—	—	1.98 s	0.008 s	3.8×10^8
90	—	—	—	0.13 s	2.5×10^{20}
360	—	—	—	16.2 s	$\sim 10^{89}$
630	—	—	—	144 s	$\sim 10^{140}$

The first three methods are all $O(\text{area}) = O(10^{2N/9})$; the tank solver is polynomial in N. Per residue N mod 9, the sequence Q admits no C-finite recurrence of order ≤ 13 and no polynomial-coefficient recurrence of order ≤ 6, degree ≤ 8 (higher orders untested) — but the *allowed* count fails identically, so base 10 lacks base 2's central-binomial collapse rather than being intrinsically wilder.

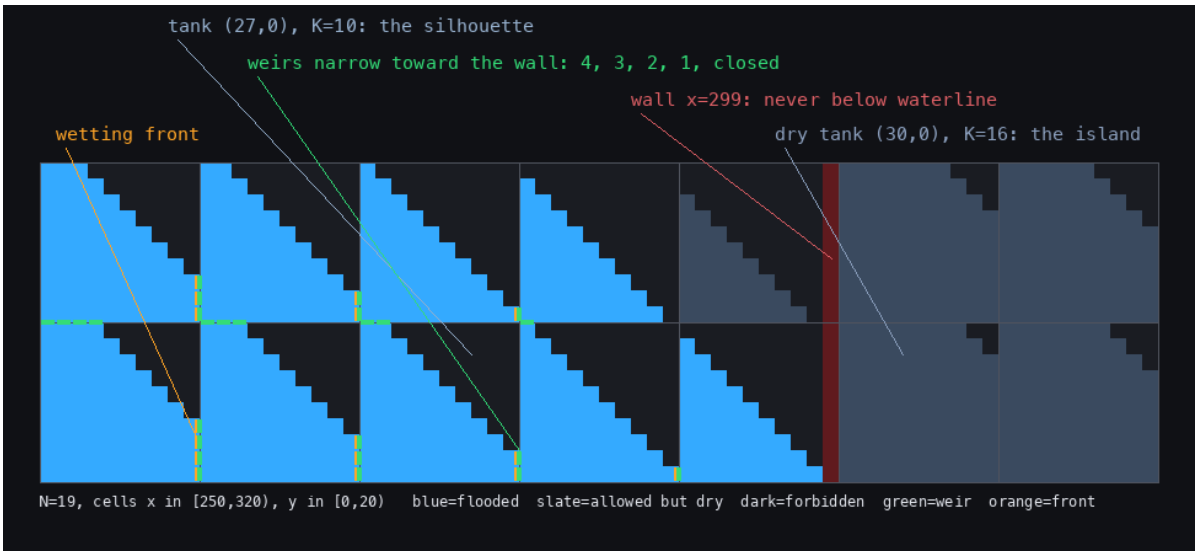


Figure 7: The glossary in one strip: cells $x \in [250, 320)$, $y \in [0, 20)$ at $N = 19$. Scale-1 tanks with budgets $K = 19 - \mathsf{s}(a) - \mathsf{s}(b)$ show their triangular silhouettes (blue: flooded; slate: allowed but dry; dark: forbidden). Weirs (green) narrow $4, 3, 2, 1$ and then close as budgets shrink toward the wall at $x = 299$ (red), whose column never dips below the waterline — so tank $(30, 0)$, despite its generous $K = 16$, stays dry: the canonical island containing $(300, 0)$. Orange marks the wetting front on each tank's exit edge.

6 The constant c: two false summits

On the *peak line* $N \equiv 8 \pmod 9$, set $c_N = \sqrt{N} \cdot$ reachable/allowed in box (the box is the k-digit square; on the peak line it exceeds the reachable bounding box only by the forbidden row and column at coordinate $10^k - 1$, digit sum $N + 1$, so the allowed counts coincide; the unrestricted allowed set is infinite, so the qualification is required).

False summit 1: $\sqrt{15}$. Pair extrapolations through $N = 629$ landed within 1.3×10^{-5} of $\sqrt{15} = 3.87298$. But exact data to $N = 989$ (tank solver, 15M states) showed c_N *cresting* at 3.86931 near $N \approx 650$ and then descending with accelerating increments — $\sqrt{15}$ exceeds every value ever attained. The agreement was a crossing artifact of two competing correction terms.

False summit 2: $c = 3.865 \pm 0.004$. With data ending at $N = 989$, model fits (calibrated on base 2, where naive $1/N$-polynomial extrapolation misses $4/\sqrt{\pi}$ by 2×10^{-2}) gave 3.865(4). The criterion of §8 later allowed evaluation at $k = 16{,}000$ ($N = 143{,}999$): the descent continues far past the old horizon, and

$$c = 3.84717(5)$$

— precisely where earlier "degenerate" power-law fits had pointed. The crest and both apparent correction families — what was *then* read as a geometric-per-decade term with ratio ~0.91 (an interpretation later withdrawn in §9) and a slow power tail — are finite-k features of one exact model. Bases $3, 4, 5$ show the same crest-then-descend shape; base 2 ($c_2 = 4/\sqrt{\pi}$, approach from above) is the degenerate case.

The summit itself. The criterion converts the peak-line count into a stay-nonnegative probability of the reflected tent walk, and the Sparre Andersen identity (Wiener–Hopf factorization in combinatorial

form) then gives the constant in closed form:

$$c_b = \frac{2\sqrt{b-1}}{\sqrt{\pi}} \exp\Big(\tfrac{1}{2}\sum_{n\geq 1}\frac{p_n}{n}\Big) = \frac{2\sqrt{b-1}}{\sqrt{\pi}} \exp\Big(-\frac{1}{4\pi}\int_{-\pi}^{\pi} \log\big(1-\varphi_b(\theta)\big)\, d\theta\Big),$$

with $p_n = \Pr(S_n{=}0)$ and $\varphi_b(\theta) = \big(\sin(b\theta/2)/(b\sin(\theta/2))\big)^2$. Base 2 collapses by hand ($\sum p_n/n = 2\log 2$, $c_2 = 4/\sqrt{\pi}$ exactly); series and integral evaluations agree to nine decimals in every base; and $c_{10} = 3.847174895\ldots$ matches the walk-DP value. The identity was verified coefficientwise against a barrier DP to 10^{-15}.

7 The spectral expedition

Instrumenting the flood recursion and normalizing tank states as (depth, d, front classes) with $d = 9k - K$ (offset of the waterline from the tank's mean elevation) yields, empirically:

1. **Exact fixed point.** The shallow census is *identical* — integer-for-integer — across all peak-line N tested (116–404), at every instrumented depth. All N-dependence lives in the deepening diffusive interior.

2. **Closure.** Transition profiles are well-defined functions of truncated front signatures (577/651 classes unambiguous; failures only at deeply submerged $d \leq -35$); classes per offset saturate at 24. The active band is one-sided: no active tank has $d > 1$.

3. **Spectrum.** The truncated transfer matrix gives $\lambda_1 \to 95.2$ (toward the trivial 100) and λ_2/λ_1 rising $0.58 \to 0.78 \to 0.86$ with truncation depth. *(The interpretation of this as approaching the observed correction ratio* ~0.91 *was later withdrawn: the* 0.91 *in c_N is a power-family fitting artifact, not an eigenvalue — see §9. The matrix gap itself remains a genuine property of the tank-count operator.)*

The d-walk steps by $(i + j) - 9$ per scale: a Markov-additive process whose continuum spectrum is the power-law correction family of §6.

8 The reachability criterion

A scalar Wiener–Hopf "warm-up" — modelling reachability as the d-walk surviving below the census barrier $d \leq 1$ — reproduced base 2's constant to 4×10^{-5} with no fitted parameters. Testing whether the model is *literally* true:

Theorem 2 (Reachability criterion; proof in [1])**.** *In base b, threshold N, $k = \lfloor N/(b{-}1)\rfloor + 1$: a quadrant cell (x, y) is reachable iff every proper digit-prefix pair (lengths $1 \leq s < k$) satisfies*

$$\mathsf{s}\big(x\langle s\rangle\big) + \mathsf{s}\big(y\langle s\rangle\big) \;\leq\; N - (b{-}1)(k{-}s) + 1,$$

and $\mathsf{s}(x) + \mathsf{s}(y) \leq N$.

Zero mismatches against exhaustive reachability across bases 3, 5, 10; integer equality with the tank solver's $Q(N)$ up to $N = 989$. The proof is elementary: *necessity* because any entry into a cell's scale-m ancestor tank crosses a trailing-nines wall, entries from the left/below certifying the *neighbor's* wall —

one digit cheaper via $\mathtt{s}(X) \le \mathtt{s}(X{-}1) + 1$, whence the +1 slack; *sufficiency* by an explicit L-path to the ancestor tank's corner — worst cells $(a{-}1)\underline{9\cdots9}$ and $a + (c{-}1)\underline{9\cdots9}$, allowed exactly under the criterion bound — followed by recursion inside the tank at threshold $N - a - c$. The two +1's coincide; the criterion is sharp. The monotone lemma follows as a corollary.

Consequences: counting reduces to a 1-D walk recursion ($O(k^2b^2)$, i.e. $O(k^2)$ for fixed base — microseconds at any N); the constant c is a one-boundary survival constant of the centered tent walk, i.e. a kernel-method/Wiener–Hopf object with an explicit (if unlovely) analytic characterization; and the 24-class front dynamics of §7, while real as 2-D geometry, is count-irrelevant — reachability depends only on ancestor budgets.

9 The $\rho \approx 0.91$ structure: a masquerade

The criterion makes c_k exactly computable, and its correction structure is a power family

$$c_k \;=\; c\left(1 + \frac{\alpha}{\sqrt{k}} + \frac{\beta}{k} + \frac{\gamma}{k^{3/2}} + \cdots\right), \qquad \alpha = \frac{1}{\sigma\sqrt{2\pi}} = 0.098213\ldots,$$

where $\sigma^2 = (b^2-1)/6$ is the variance of a tent step, with α forced by the denominator alone: the lattice atom $\Pr(S_k{=}0)$ in the allowed-count normalization $(1-p_k)/2$ ($1/\sqrt{\pi}$ exactly in base 2). The numerator contributes no $k^{-1/2}$ term: writing $w = 1 - z$ and $G(z) = \frac{1}{2}\sum_n p_n z^n/n$, the Sparre Andersen generating function of the stay-nonnegative probabilities q_k is $w^{-1/2}e^{G(z)}$, and since $p_n \sim \alpha n^{-1/2}$ the leading singular part of G is $-\alpha\sqrt{\pi}\,w^{1/2}$, whose product with the prefactor is the analytic constant $w^{-1/2}\cdot w^{1/2} = 1$, contributing nothing to $[z^k]$.

Pushed one order further, the same expansion gives β in closed form (`digitsum_beta.py`). Let $K = e^{G(1)}$, so that $c_b = 2\sqrt{b-1}\,K/\sqrt{\pi}$, and $\rho_1 = \sum_{n\ge1}\big(p_n - \alpha n^{-1/2}\big)$, convergent since $p_n - \alpha n^{-1/2} = O(n^{-3/2})$. Expanding $\mathrm{Li}_{3/2}$ at $z = 1$ gives $G(z) = \log K - \alpha\sqrt{\pi}\,w^{1/2} - \frac{1}{2}\big(\alpha\,\zeta(\frac{1}{2}) + \rho_1\big)w + O(w^{3/2})$, and singularity analysis of $w^{-1/2}e^G$ yields $q_k = K(\pi k)^{-1/2}\big(1 + u_1/k + O(k^{-2})\big)$ with $u_1 = -\frac{1}{8} + \frac{1}{4}\big(\alpha\,\zeta(\frac{1}{2}) + \rho_1\big) - \frac{1}{4}\pi\alpha^2$. Reachability needs endpoint ≥ 1, so the endpoint atom $r_k = \Pr(S_1, \ldots, S_{k-1} \ge 0,\ S_k = 0)$ is subtracted; by the Spitzer–Baxter identity its generating function is the u^0 coefficient of $\exp\sum_n \frac{z^n}{n}\,\mathbb{E}[u^{S_n}; S_n \ge 0]$, namely $e^{2G(z)}$ (verified coefficientwise to 10^{-17}; $C_{k+1}/4^k$ in base 2), so $r_k \sim K^2\alpha\,k^{-3/2}$. Assembling $c_k = 2\sqrt{(b-1)k-1}\,(q_k - r_k)/(1-p_k)$ with $K\alpha\sqrt{\pi} = K/(\sigma\sqrt{2})$,

$$\beta \;=\; \Big(1 - \frac{\pi}{4}\Big)\alpha^2 - \frac{K}{\sigma\sqrt{2}} - \frac{1}{8} - \frac{1}{2(b-1)} + \frac{\alpha\,\zeta(\frac{1}{2}) + \rho_1}{4}.$$

In base 2 everything is explicit ($K = 2$, $G = \log\big(2/(1+\sqrt{w})\big)$, so $\alpha\zeta(\frac{1}{2}) + \rho_1 = -1$): $\beta_2 = 1/\pi - 25/8 = -2.80669\ldots$. In base 10 the formula gives $\beta = -0.414716$, and the power family fitted to the exact c_k on $k \in [300, 1500]$ with α pinned returns -0.414716; with α and β both pinned, two fitted higher terms reproduce c_k to rms 5×10^{-11} and extrapolate out-of-window to 3.5×10^{-6}. No geometric term is needed, and any admixture is bounded far below the historical claim. Three confirmations: the crest is the two-term power phenomenon $k^* = (2|\beta|/\alpha)^2 = 71$ (observed 73, $N = 656$); refitting a single geometric on the historical window $q \in [28, 69]$ returns $\rho = 0.9153$ at rms 1.6×10^{-5} — precisely the numbers that launched the "subleading eigenvalue" reading, now identified as arithmetic rather than spectroscopy; and the *true* subleading singularity of the Sparre Andersen generating function, from the first side lobe of the tent kernel, has ratio $\varphi^* = 0.0505$ per step (singularity at $z \approx 19.8$) — a genuine geometric far too small to ever matter. The λ_2/λ_1 reading of §7 is accordingly withdrawn; in hindsight the exactly stationary shallow census already said the convergence was not geometric.

10 From expedition to proof

This report stops deliberately at the *statement* of the criterion (§8). Its proof, and the corollaries that follow from it, are the subject of the companion paper [1], which is written in the conventional order — setting, theorem, necessity, sufficiency, corollaries — that the expedition did not follow. The two documents are meant to be read together, and the correspondence between them is close enough to tabulate: every structure met along the way reappears in [1] as a consequence of the two-line test.

This report	Companion paper [1]
Axis cliffs and the staircase in N (§2)	the budget inequality at the leading digit
Monotone lemma (§3)	Corollary (Monotone-staircase lemma)
Base-2 closed form $\binom{2N+3}{N}$ (§4)	Corollary (Base 2): reflection and the surfacing map
Gastank solver (§5)	Corollary (Exact counting): the tent-weight walk
The constant c (§6)	Corollary (Asymptotics): the Wiener–Hopf constant
Spectral reading of ρ (§7, §9)	no counterpart: a finite-window artifact

Read down the right-hand column, the proof paper is short because the criterion does the work once. The cliffs at multiples of 9 are the budget inequality evaluated at the leading digit, where one unit of slack decides the fate of a whole column. The monotone lemma, found by experiment and proved here by a local argument, becomes the statement that undirected, right/up, and criterion reachability coincide. The gastank recursion — the first polynomial-time solver, arrived at by watching floods fill self-similar tanks — is the one-dimensional walk over tent weights: the tank walls are the digit positions at which the walk is checkpointed, and the carry-free property is the reason the recursion closes. The constant c, which two false summits placed at $\sqrt{15}$ and then at a subleading eigenvalue, is the Wiener–Hopf factorization constant of the tent kernel, exact in base 2 and 3.847174895 . . . in base 10; the brute-force fits of §6 reached five of those digits. Only the spectral reading has no counterpart, because there was nothing to prove: §9 shows the apparent ratio ∼0.91 to be what a power series looks like through a window of nine decades.

What the proof does not settle is also worth recording. For $b \geq 3$ no closed form for $Q(N)$ is known; the search of §5 found no low-order C-finite or polynomial-coefficient recurrence per residue class, but nonexistence is not proved, and the transcendental-looking form of c_b suggests none should be expected. The correction family of §9 is now derived through order $1/k$ — α from the lattice atom in the denominator, β from the Sparre Andersen and Spitzer–Baxter expansions — and only the coefficients from $k^{-3/2}$ on remain fitted. And the base-10 sequence $f(N)$ is not yet in the OEIS, although its base-2 quadrant cousin has been there for decades as A002054; a submission is prepared in the project repository.

The order of presentation was a choice. Written theorem-first, this material is five pages; written as it happened, it is nine, and two of them are wrong. We have kept the longer version because the false summits were not detours from the criterion but the route to it: the $\sqrt{15}$ fit forced the precision that exposed the correction family, and the eigenvalue reading forced the spectral analysis whose failure made the Markov-additive structure — and hence the criterion — visible.

References

[1] A. Benzan, *An exact reachability criterion for digit-sum–bounded lattice walks*, companion paper, July 2026. Distributed with this report as `criterion.pdf` in the project repository.